

Local AI Orchestration Platform

Final Year Project – Mid Term Evaluation Document

Problem Statement

Organisations in privacy-sensitive domains such as healthcare, legal, and finance cannot adopt cloud-based AI platforms like ChatGPT, Claude, or Gemini due to data residency regulations and compliance requirements (HIPAA, GDPR). Sending patient records, legal documents, or financial data to third-party servers exposes organisations to legal liability and data breach risk.

The existing alternative, deploying a single large language model locally, is computationally and economically infeasible for most organisations. State-of-the-art LLMs require significant GPU infrastructure and do not natively cover multimodal tasks such as document retrieval, speech interaction, or computer vision.

Proposed Solution

A local, offline, multi-pipeline AI orchestration system that can be configured and deployed by organisations within their own infrastructure. A central orchestrator routes user queries to the most appropriate specialised pipeline, rather than relying on a single large generalist model. The system is designed to be domain-configurable, meaning the same platform can be tuned and deployed for healthcare, legal, software, or any other domain via configuration rather than re-engineering.

Core Pipelines (Subject to change)

- **RAG Pipeline:** Document ingestion, OCR, vector embeddings, and retrieval-augmented question answering over organisation-owned documents.
- **Fine-tuned LLM:** A smaller language model fine-tuned on domain-specific data, replacing the need for a large generalist model at a fraction of the compute cost.
- **Speech Pipeline (STT/TTS):** Speech-to-text and text-to-speech for voice-based interaction, enabling accessibility and hands-free workflows.
- **Computer Vision Pipeline:** Image and visual document understanding for tasks such as report parsing, scan analysis, and visual query answering.
- **Orchestrator:** A central routing layer that classifies incoming queries and dispatches them to the appropriate pipeline, with a single unified user interface.

Memory and Knowledge Graph Layer

A persistent, local memory layer built on a knowledge graph that accumulates organisational context over time as the platform is used. This includes relationships between departments, personnel, workflows, cases, and documents. Unlike cloud platforms, there is no external limit on how much context can be retained or how sensitive the stored information can be, because all data remains within the organisation's own infrastructure.

Key Novelties

- **Multi-pipeline over single-model:** Routing queries to small specialised models is more computationally feasible and often more accurate than running one large model for all tasks.
- **Configuration-driven domain adaptability:** The same system can be redeployed across different domains by swapping domain-specific configuration files, without code changes.
- **Persistent organisational memory:** Knowledge graph-based memory that builds a representation of the organisation over time, which no cloud platform can offer due to privacy and regulatory constraints on data retention.
- **Fully local and offline:** No data leaves the organisation's infrastructure at any point, making it viable for HIPAA, GDPR, and similar compliance environments.

Research Gap Being Addressed

Existing literature addresses privacy-preserving local deployment, multi-agent orchestration, and LLM-knowledge graph integration independently. No existing system combines all three in a unified, domain-configurable, production-ready platform. This project addresses that gap.

Points to Defend at Evaluation

- Why not just run a local LLaMA or Mistral model? (Answer: single LLM lacks RAG, vision, speech, and persistent memory; requires expertise to configure per domain; our system provides all of this orchestrated and configurable out of the box)
- What is the actual compute requirement for your system vs a single large LLM? (Have a rough comparison ready)
- Which specific compliance regulations make cloud AI unusable for your target domains? (HIPAA for healthcare, legal privilege for legal, GDPR broadly)
- What is novel about the memory layer vs just a database? (Answer: knowledge graph captures relationships and context, not just raw data; it grows richer as the platform is used; no external retention cap)
- How does the orchestrator decide which pipeline to route a query to? (Classifier model or rule-based routing – be specific about your approach)
- What does domain-configurable mean concretely? (Domain-specific config files: prompts, model weights, data paths, retrieval corpora – no code changes required)
- Which research papers directly back your architecture choices? (HuggingGPT, TaskMatrix.AI for multi-pipeline; LLM-KG integration papers for memory layer; privacy LLM paper for compliance framing)
- What is your evaluation metric for success? (Define at least one per pipeline: RAG accuracy, LLM task performance, STT WER, memory recall precision)
- Who are your competitors and what do they not do that you do? (Azure on-prem AI, private cloud LLM deployments – none combine all pipelines with persistent org memory in one configurable system)
- What is the minimum viable hardware for your target organisation to deploy this? (Have a realistic spec ready)